12.7 Who Meets Whom in a Knockout

WORKSHEET

The question

STEP II Specimen paper, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q16

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How many matches are played in a knockout tournament with 2^n players? (Each match eliminates one player.)

STAGE 1 The first round

Covers: Part (i)

Ask yourself In the first round, the first player's opponent is equally likely to be any of the other players. How many are there?

STAGE 2 The key symmetry

Covers: Part (ii)

Ask yourself Every draw is random and every match is a fair coin toss. Does the process treat any pair of players differently from any other? So, for any one particular match, how likely is it to be between your two players?

STAGE 3 The final

Covers: Part (ii)

Pause and try The final is one match, so use the second stage. Round r contains 2^{n-r} matches, and two players can meet at most once in a round: so what is the probability that they meet in round r?

STAGE 4 **At any stage**

Covers: Part (iii)

Ask yourself They can meet at most once, so the events "meet in round r" are mutually exclusive. Add them up, or count all $2^n - 1$ matches at once.

What's really going on

Expected counts

The argument in Stage 4 is really an expectation calculation: the number of times X and Y meet is a sum of indicators, one per match, and its expectation (which equals the probability of meeting, since it is 0 or 1) is the sum of the indicators' expectations. This is linearity of expectation, used in Section 9.11 and Section 12.4. Its great advantage is that it ignores all the complicated dependence between the rounds.

Seeded tournaments

Real tournaments are *seeded*: the strongest players are placed so that they cannot meet until the late rounds. And real matches are not 50–50. Both destroy the symmetry, and calculating the chance that two given players meet becomes a genuine computation. Statisticians who model sport use exactly such calculations, with each match's probability estimated from players' ratings, to forecast tournaments. The symmetric case is the baseline against which the real effects are measured.

Try this next

1. **Round by round.** Show directly that the probability that X and Y meet in round k is $\frac{2^{n-k}}{\binom{2^n}{2}}$, and check that these add up to $\frac{1}{2^{n-1}}$.
2. **A fixed bracket.** If instead the bracket is fixed in advance (as in the figure) but all players are placed in it at random, does the answer change?
3. **A strong player.** Suppose player X wins every match. What is now the probability that X meets Y?

4 **A league.** In a league of $2n$ teams where every pair meets once, how many matches are there? What is the analogous "probability of meeting"?

Further reading

- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover), which includes tournament problems.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on modelling and forecasting.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online), on indicators and symmetry.

The full guided solution is in Section 12.7:
https://beyondstep.keithmcnulty.org/ch12/knockout.html